

Wazuh SIEM Home Lab Portfolio Documentation

Project Title: Wazuh SIEM Deployment and Security Monitoring Lab

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Environment: Home Lab Virtual Environment

Project Overview

This project demonstrates the deployment and configuration of a Security Information and Event Management (SIEM) platform using **Wazuh** to monitor endpoint activity across Windows and Linux systems. The objective of this lab was to simulate real-world security events and analyze them through centralized logging and alerting.

The environment consisted of:

- **Wazuh SIEM Server**
- **Windows Virtual Machine (Target Endpoint)**
- **Kali Linux Virtual Machine (Attacker Machine)**

Multiple security scenarios were simulated, including authentication failures, malware detection, brute-force attacks, file integrity monitoring, and PowerShell activity monitoring.

This project demonstrates hands-on experience in:

- SIEM deployment
 - Endpoint monitoring
 - Threat detection
 - Log analysis
 - Security event investigation
 - Basic threat simulation
-

Lab Environment Setup

Infrastructure Components

System	Role	Description
Wazuh Server	SIEM	Centralized log collection and monitoring
Windows VM	Endpoint	Target system monitored by Wazuh agent
Kali Linux VM	Attacker	Used to simulate attacks
Proxmox	Hypervisor	Virtual machine hosting platform using a 3 node cluster.

Network Diagram (Simple Version)

You can recreate this visually later:

```
graph TD;
  Kali[Kali Linux (Attacker)] --- Windows[Windows VM (Target)];
  Windows --- Wazuh[Wazuh Server];
  Wazuh --- Dashboard[Wazuh Dashboard];
```

(Optional future improvement: create this in **draw.io**)

Lab 1 — Windows Authentication Monitoring

Objective

To validate that Windows authentication events are successfully collected and monitored by the Wazuh SIEM.

Tools Used

- Windows Event Logs
 - Wazuh Agent
 - Wazuh Dashboard
-

Procedure

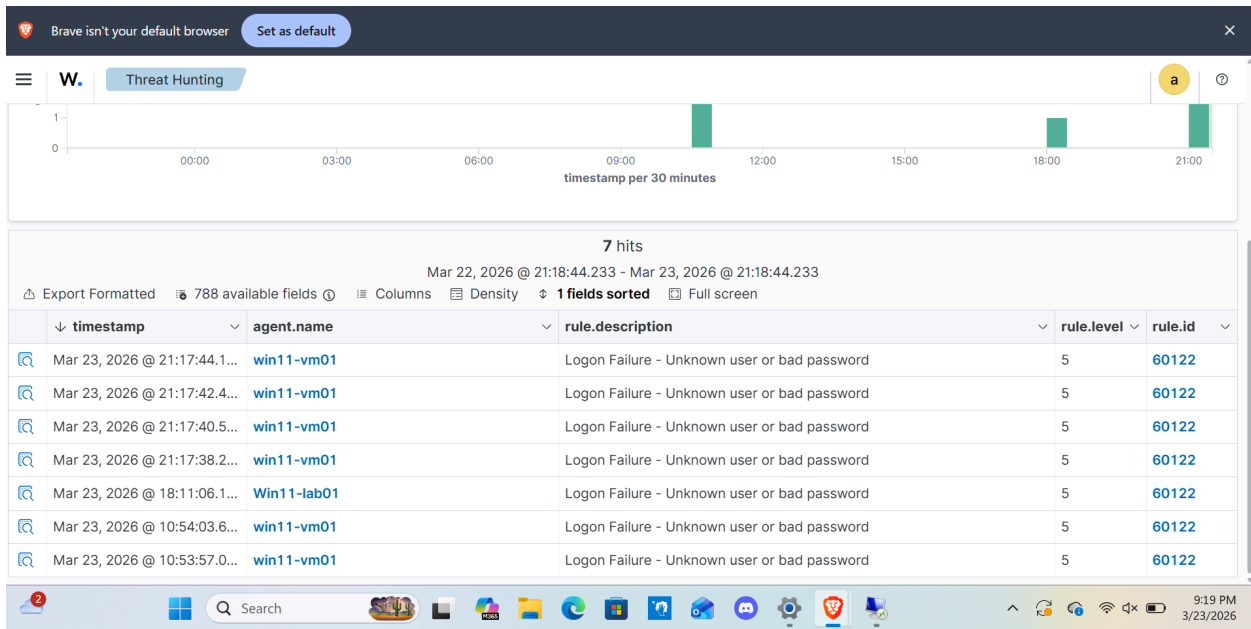
1. Locked the Windows system.
 2. Entered incorrect password attempts multiple times.
 3. Successfully logged in afterward.
 4. Navigated to Wazuh Dashboard.
 5. Viewed authentication logs.
-

Events Observed

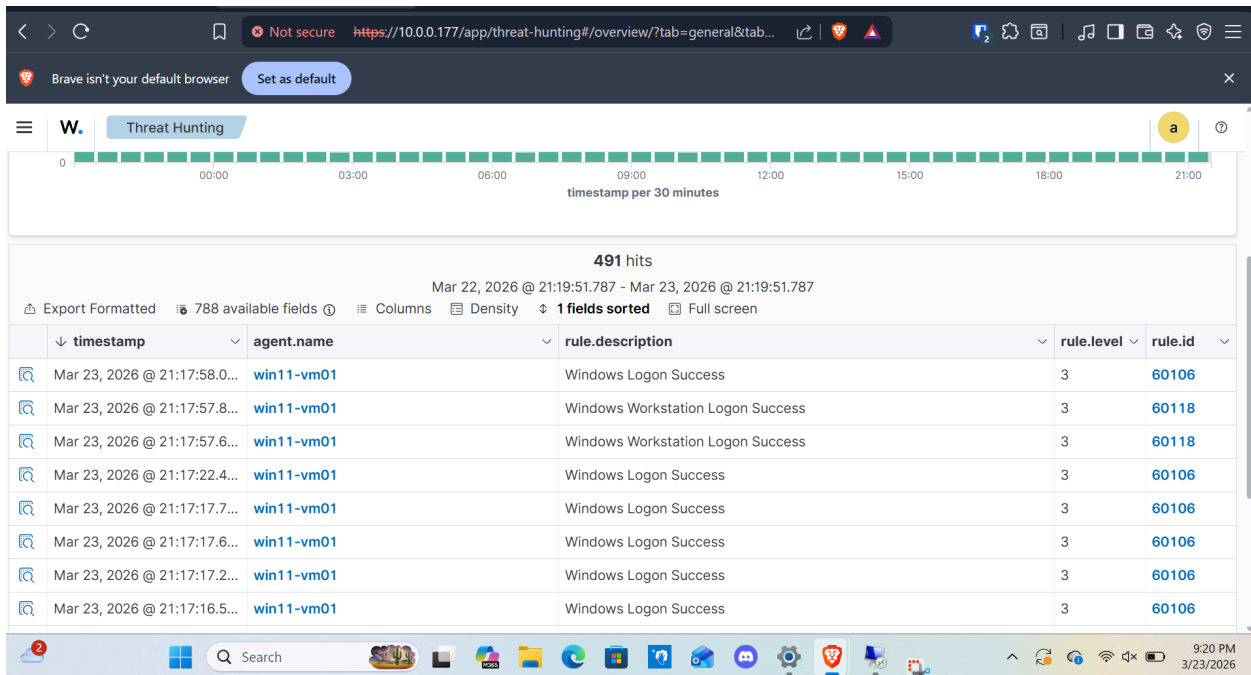
Event ID	Description
4625	Failed login attempt
4624	Successful login

Screenshots to Include

Failed login alert:



Successful login alert:



Authentication dashboard view:

Tools Used

- Windows Defender
 - EICAR test file
 - Wazuh SIEM
-

Procedure

Pre: Added the firewall logs to Wazuh alerts

```
<localfile>  
  <location>Microsoft-Windows-Windows Defender/Operational</location>  
  <log_format>eventchannel</log_format>  
</localfile>
```

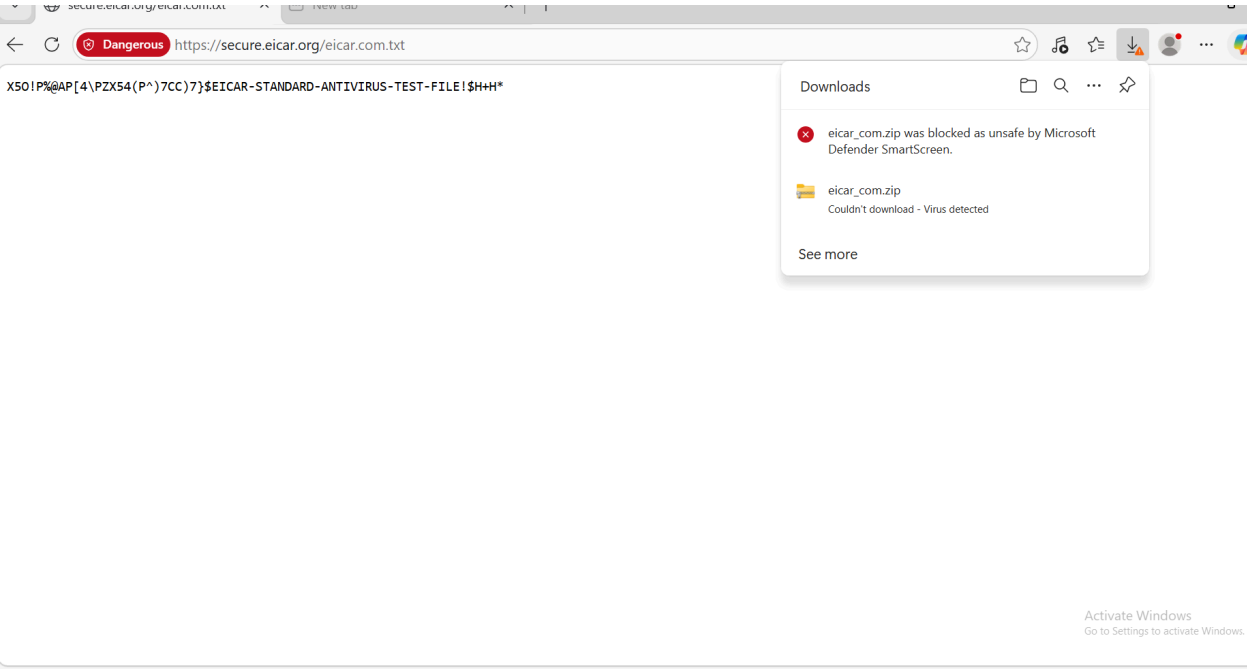
1. Downloaded EICAR test file.
 2. Saved file to Windows desktop.
 3. Observed endpoint response.
 4. Monitored Wazuh alerts.
-

Alerts Observed

- Malware detection alert
 - File creation event
 - Windows Defender activity
-

Screenshots to Include

EICAR file download



Wazuh malware alert:

timestamp	agent.name	rule.description	rule.level	rule.id
Mar 23, 2026 @ 21:41:10.8...	win11-vm01	Windows Defender: Antimalware platform performed an action to protect ...	3	62124
Mar 23, 2026 @ 21:41:05.7...	win11-vm01	Windows Defender: Antimalware platform detected potentially unwanted ...	12	62123
Mar 23, 2026 @ 21:40:38.2...	Win11-lab01	Windows Logon Success	3	60106
Mar 23, 2026 @ 21:40:32.4...	Win11-lab01	Windows Logon Success	3	60106
Mar 23, 2026 @ 21:40:23.1...	win11-vm01	Wazuh agent started.	3	503
Mar 23, 2026 @ 21:40:13.2...	win11-vm01	Wazuh agent stopped.	3	506
Mar 23, 2026 @ 21:38:59.1...	win11-vm01	Windows Logon Success	3	60106
Mar 23, 2026 @ 21:38:41.6...	win11-vm01	Windows Logon Success	3	60106
Mar 23, 2026 @ 21:35:54.5...	win11-vm01	Service startup type was changed	3	61104

File detection log

Outcome

The EICAR test file successfully triggered malware alerts within the SIEM, confirming endpoint threat detection capabilities.

Skills Demonstrated

- ✓ Malware simulation
 - ✓ Threat detection
 - ✓ Endpoint monitoring
-

Lab 3 — Brute Force Attack Simulation

Objective

To simulate a brute-force login attack using Kali Linux and observe detection through Wazuh.

Tools Used

- Kali Linux
 - Hydra
 - Windows RDP
 - Wazuh SIEM
-

Procedure

1. Identified Windows target IP address.
 2. Used Hydra to simulate brute-force login attempts.
 3. Monitored Wazuh authentication logs.
 4. Observed repeated failed login attempts.
-

Command Used

```
hydra -t 4 -V -f -l administrator -P /usr/share/wordlists/rockyou.txt rdp://WINDOWS_IP
```

Alerts Observed

- Multiple failed login attempts
 - Authentication failure logs
 - Brute-force pattern behavior
-

Screenshots to Include

 Hydra execution

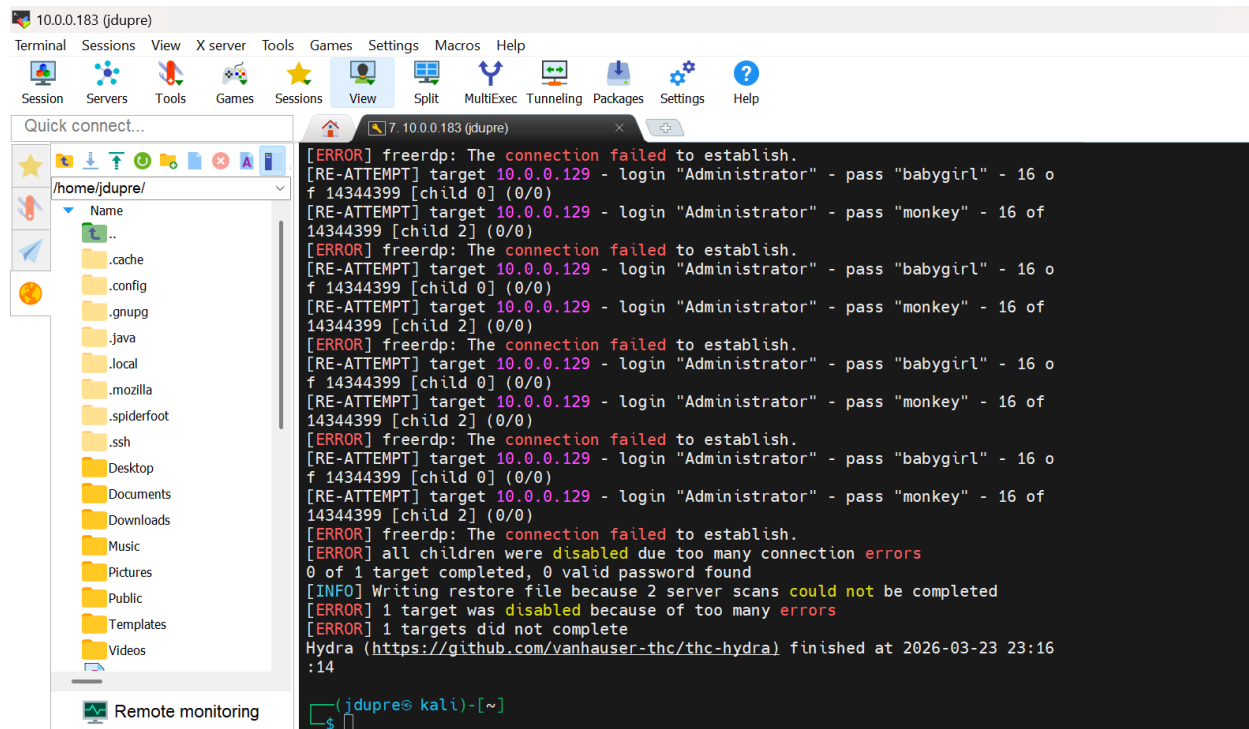
```
Fetches 75.8 MB in 9s (8,532 kB/s)
1114 packages can be upgraded. Run 'apt list --upgradable' to see them.

(jdupre@kali)-[~]
$ sudo apt install hydra -y
hydra is already the newest version (9.6-3).
The following packages were automatically installed and are no longer required:
gcc-14-base libgpgmepp6t64
gir1.2-girepository-2.0 libobjc-14-dev
libcrypt-dev libstdc++-14-dev
libdisplay-info2 libwireshark18
libgcc-14-dev libwiretap15
libgirepository-1.0-1 libwsutil16
libgpgme11t64 linux-image-6.12.38+kali-amd64
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1114

(jdupre@kali)-[~]
$
```

```
(jdupre@kali)-[~]  
$ hydra -t 4 -V -l Administrator -P /usr/share/wordlists/rockyou.txt rdp://10.0.0.129  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-23 23:16:06  
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task  
[DATA] attacking rdp://10.0.0.129:3389/  
[ATTEMPT] target 10.0.0.129 - login "Administrator" - pass "123456" - 1 of 1434399 [child 0] (0/0)  
[ATTEMPT] target 10.0.0.129 - login "Administrator" - pass "12345" - 2 of 1434399 [child 1] (0/0)  
[ATTEMPT] target 10.0.0.129 - login "Administrator" - pass "123456789" - 3 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.0.0.129 - login "Administrator" - pass "password" - 4 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.0.0.129 - login "Administrator" - pass "iloveyou" - 5 of 14344399 [child 0] (0/0)  
[ATTEMPT] target 10.0.0.129 - login "Administrator" - pass "princess" - 6 of 14344399 [child 2] (0/0)
```



Wazuh brute-force logs:

W. Threat Hunting

timestamp per 30 minutes

21 hits

Mar 22, 2026 @ 22:17:00.112 - Mar 23, 2026 @ 22:17:00.113

Export Formatted 788 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Mar 23, 2026 @ 22:16:13.1...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:13.1...	win11-vm01	User account locked out (multiple login errors)	9	60115
Mar 23, 2026 @ 22:16:13.1...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:13.1...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:13.1...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:11.1...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:11.1...	win11-vm01	Multiple Windows Logon Failures	10	60204
Mar 23, 2026 @ 22:16:11.1...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:11.1...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:09.0...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:09.0...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:09.0...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:09.0...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:09.0...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 22:16:09.0...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122
Mar 23, 2026 @ 21:17:44.1...	win11-vm01	Logon Failure - Unknown user or bad password	5	60122

Rows per page: 15

10:20 PM

Outcome

Multiple failed login attempts were successfully logged and detected by Wazuh, validating brute-force detection capabilities.

Skills Demonstrated

- ✓ Attack simulation
 - ✓ Brute-force detection
 - ✓ Security log investigation
-

Lab 4 — PowerShell Activity Monitoring

Objective

To monitor command execution activity using PowerShell.

Tools Used

- Windows PowerShell
 - Wazuh SIEM
-

Procedure

1. Opened PowerShell.
2. Executed command:

whoami

Get-Process

Get-Service

powershell -ExecutionPolicy Bypass

Invoke-WebRequest -Uri http://example.com -OutFile test.txt

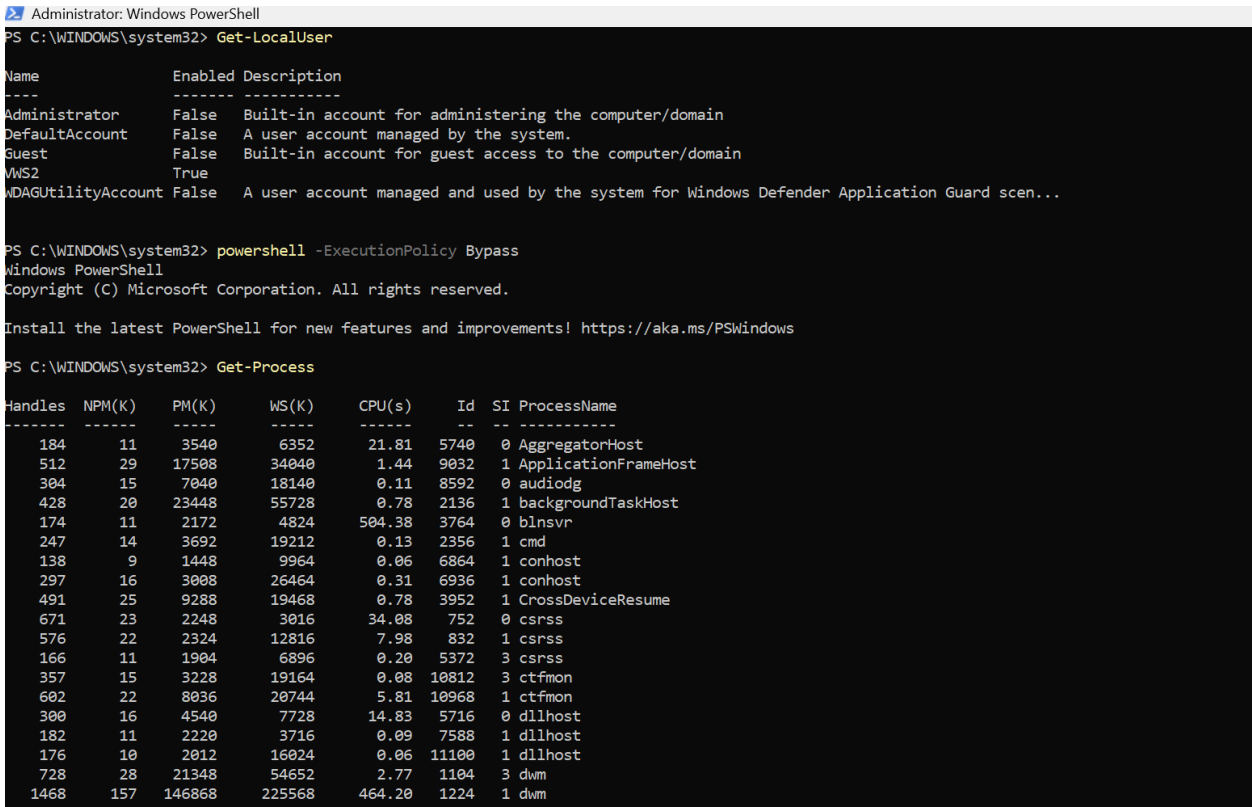
3. Monitored Wazuh logs.
-

Events Observed

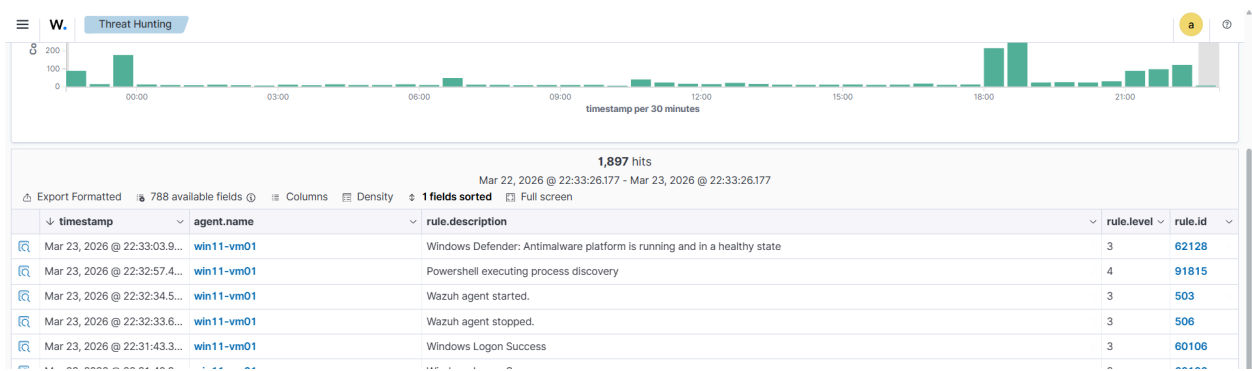
- PowerShell execution logs
- Command execution visibility

Screenshots to Include

PowerShell command



Wazuh command logs



Outcome

PowerShell activity was successfully logged and displayed within the SIEM environment.

Skills Demonstrated

- ✓ Command monitoring
 - ✓ Activity logging
 - ✓ Security visibility
-

Lab 5 — File Integrity Monitoring (FIM)

Objective

To verify that file changes are monitored and logged.

Tools Used

- Windows File System
 - Wazuh File Integrity Monitoring
-

Procedure

1. Created file:

C:\testfile.txt

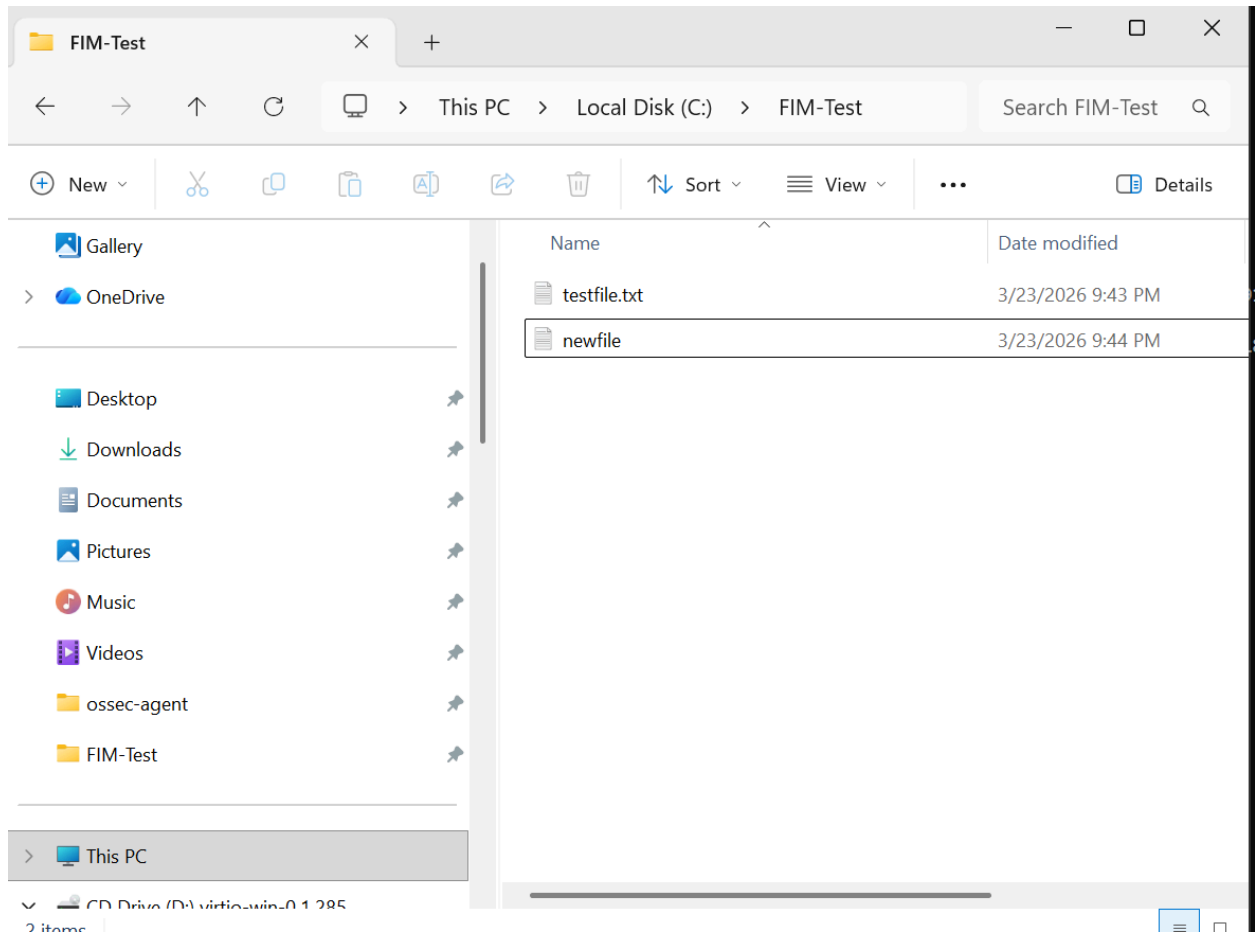
2. Modified file multiple times.
 3. Observed file change logs.
-

Events Observed

- File creation
- File modification
- File change detection

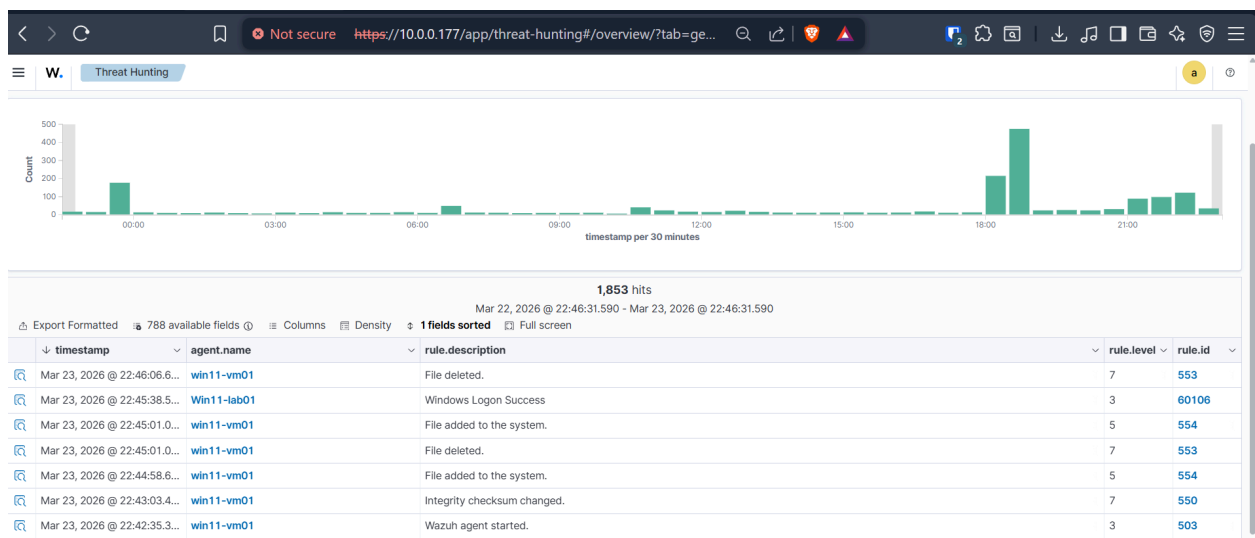
Screenshots to Include

 File creation	
NAME	Date Modified
 FIM-Test	3/23/2026 9:40 PM
 inetpub	9/15/2025 12:47 PM
 PerfLogs	4/1/2024 12:26 AM
 Program Files	3/8/2026 9:36 PM
 Program Files (x86)	3/8/2026 10:14 PM
 Users	3/8/2026 9:27 PM
 Windows	3/10/2026 5:32 PM
 testfile.txt	3/23/2026 9:43 PM



File was deleted.

Wazuh FIM alerts:



Outcome

File modifications were successfully logged, demonstrating effective file integrity monitoring.

Skills Demonstrated

- ✓ File monitoring
 - ✓ Integrity detection
 - ✓ Endpoint logging
-